Faculty of Science, Engineering and Built Environment

SEE210 Power Engineering Design

Deakin University Unit Guide

Trimester 1, 2025

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Welcome

Power engineering design is the design of electrical systems for the supply and distribution of electrical power through a network to end user for their needs. The primary goal is to generate the required amount of power and distribute it to where it's required, whilst doing this in a safe, reliable, and efficient manner. In this unit, students will engage in a Power Engineering Design Project individually and as a team. Students will investigate different power sources, transformers, transmission, and distribution lines to address stakeholder needs and develop a system design by applying ethical and Australian OHS standards. This unit lays the foundations for understanding and designing modern power systems network comprised of conventional and renewable energy sources. Students will develop knowledge and technical competence in finding power engineering solutions to a range of electrical engineering problem scenarios.
This Unit Guide provides you with the key information about this unit. Please read it carefully and refer to it frequently throughout the study period. Your unit site also provides information about your rights and responsibilities. We will assume you have read this before the unit commences, and we expect you to refer to it throughout the study period.

To be successful in this unit, you must:

- read all materials in preparation for your learning activities and follow up each with further study and research on the topic
- start your assessment tasks well ahead of the due date
- read or listen to all feedback carefully and use it in your future work
- attend and engage in all educator facilitated (scheduled) learning activities and other learning experiences as part of the unit design

Who is the unit team?

Unit chair: leads the teaching team and is responsible for overall delivery of this unit

Name: Shama Islam
Campus: Geelong Waurn Ponds
Email: shama.i@deakin.edu.au
Phone: +61 3 522 78139
Other members of the team and how to contact them

There are no other staff involved in teaching this unit.
Administrative queries

- check-out the 'SEBE Student Hub' section on your unit site
- contact your Unit Chair or Campus Leader
- drop in or contact Student Central to speak with a Student Adviser

For additional support information, please see the Rights and Responsibilities section under 'Content' in your unit site.
About this unit

Unit development in response to student feedback

Every trimester, we ask students to tell us, through eVALUate, what helped and hindered their learning in each Unit. You are strongly encouraged to provide constructive feedback for this Unit when eVALUate opens (you will be emailed a link).
In previous versions of this unit, students have told us that these aspects of the Unit have helped them to achieve the learning outcomes

- Hands-on studio activities.
- Interactive seminars.

The following aspects of the unit have been introduced, enhanced or retained in response to feedback from students who have undertaken this unit in previous trimesters:

- Improved lecture recordings.
- More discussion on teamwork expectations.

If you have any concerns about the Unit during the trimester, please contact the unit teaching team - preferably early in the trimester - so we can discuss your concerns, and make adjustments, if appropriate.

Learning Outcomes

Each Unit in your course is a building block towards Deakin's Graduate Learning Outcomes - not all Units develop and assess every Graduate Learning Outcome (GLO).

ULO	These are the Unit Learning Outcomes (ULOs) for this unit. At the completion of this unit, successful students can:	Alignment to Deakin Graduate Learning Outcomes (GLOs)
ULO1	Apply specialised knowledge and technical competence in proposing engineering solutions to a range of electrical engineering problem scenarios.	GLO1: Discipline-specific knowledge and capabilities GLO4: Critical thinking GLO5: Problem solving
ULO2	Collaborate as a member of a team to learn research, evaluate, and use information effectively to propose solutions to problem scenarios.	GLO2: Communication GLO4: Critical thinking GLO7: Teamwork
ULO3	Communicate project details through the use of oral and written communication to a professional audience.	GLO2: Communication
ULO4	Integrate Occupational Health and Safety (OHS) and work responsibly and safely in engineering environments to demonstrate professionalism. (practical)	GLO6: Self-management
ULO5	Demonstrate transferrable employability skills by working collaboratively in an engineering team to develop, implement and complete a project management strategy for evaluating authentic real world stakeholders needs including indigenous perspectives.	GLO1: Discipline-specific knowledge and capabilities GLO2: Communication GLO4: Critical thinking GLO7: Teamwork GLO8 Global Citizenship

Assessing your achievement of the unit learning outcomes

Hurdle requirements

To be eligible to obtain a pass in this unit, students must achieve a minimum of 50% in the project validation presentation assessment task.

Brief summary of the hurdle requirement	Rationale
To be eligible to obtain a pass in this unit, students must achieve a minimum of 50% in the project validation presentation assessment task.	In accordance with external accreditation requirements, the School of Engineering incorporates a hurdle requirement in order to achieve the Unit Learning Outcomes (ULOs) and Course Learning Outcomes (CLOs) of the course.

Summative assessment (tasks that will be graded or marked)

NOTE: It is *your responsibility* to keep a backup copy of every assignment and the materials used to develop/complete it where possible (e.g. written/digital reports, essays, videos, images). In the unusual event that one of your submissions becomes corrupted, is incorrectly submitted or otherwise lost, you may be asked to submit the backup copy. Any work you submit may be checked by electronic or other means for the purposes of detecting breaches of academic integrity such as collusion, plagiarism and contract cheating. You must understand your responsibility to act with honesty and integrity in your studies as Deakin takes all breaches very seriously. Make sure you read Your rights and responsibilities as a student in this unit to find out more about academic integrity.

Deakin has a universal assessment submission time of 8 pm AEDT/AEST. A late penalty will apply to assessments submitted after 11.59 pm AEDT/AEST.

Summative assessment task 1

Title	**Project Design Plan (group)**
Brief description of assessment task	This assessment task requires students to work in a team to develop a project design in power system networks, including developing equipment requirements, timeframe and deliverables as well as OHS considerations in undertaking the practical work. Group strengths and weaknesses, plans for communication and team management will also be addressed. In addition, real world stakeholder needs including indigenous perspectives will be considered. Through this assessment task, the students will develop teamwork and project management skills.
Detail of student output	This is a group task. Each team will submit a Project Design Plan that is not more than 6 pages in total. Students will present a technical summary of the project detailing aims, scope, assumptions, constraints and a project plan to complete the design in a timely and effective manner. In addition, students will complete a peer-feedback literacy module which will enable them to provide feedback on teamwork effectively.
Grading and weighting (% total mark for unit)	15% - marked and graded
This task assesses your achievement of these Unit Learning Outcome(s)	ULO1 - Through student ability to develop solutions for real-world power engineering problem scenarios. ULO2 - Through student ability to collaborate with team members in planning a power engineering project. ULO3 - Through student ability to communicate a detailed plan for delivering a real-world power engineering project. ULO5 - Through student ability to interpret stakeholder requirements and deliver effective project management strategies in a team environment.
This task assesses your achievement of these Graduate Learning Outcome(s)	GLO1 - Through student ability to demonstrate their understanding of the key concepts of engineering professional practice relevant to power engineering design. GLO2 - Through student ability to communicate their ideas in a team environment through a written document to a professional audience. GLO4 - Through student ability to apply analytical skills to develop a detailed project design plan. GLO5 - Through student ability to propose solutions for designing a small and practical power system. GLO7 - Through student ability to collaborate with members of a team to develop a detailed project design plan. GLO8 - Through student ability to apply ethical considerations for project management including indigenous perspectives.
How and when you will receive feedback on your work	Students will receive feedback on their plan from their facilitator during the online and on-campus seminars.
When and how to submit your work	Week 5, Thursday 3 April 2025 by 8pm (AEST), via the unit site.

Summative assessment task 2

Title	**Design Problem Solving Tasks x 2**
Brief description of assessment task	These assessment tasks require students to work independently to solve a series of design problems. The problems will require both written responses and mathematical calculations. Students will apply their understanding of the content covered in Power Engineering Design. There will be two design problem solving tasks, within this unit, one at the mid and one towards the end of the trimester.

Detail of student output	These are individual tasks. Students will receive the design problem solving task through the unit site and will upload their answers into quiz platform on the unit site within a 2-hour timeframe. Multiple 2-hour time slots will be available for students to choose from for completing the task. Students will be applying their problem-solving skills and understanding of theoretical concepts related to power system design.
Grading and weighting (% total mark for unit)	30% (2 x 15%) - marked and graded
This task assesses your achievement of these Unit Learning Outcome(s)	ULO1 - Through student ability to apply technical understanding of power engineering concepts in solving real-world power engineering design problems.
This task assesses your achievement of these Graduate Learning Outcome(s)	GLO1 – Through student ability to demonstrate their understanding of the key concepts of power engineering design calculations.
How and when you will receive feedback on your work	Students will receive feedback during the online seminars. Students will work through design problems that will be similar to those they will encounter during the assessment task.
When and how to submit your work	Week 6: Wednesday 9 April 2025, 8pm AEST* (Design Problem Solving Task 1) Week 12 (Study Period): Wednesday 28 May 2025, 8pm AEST* (Design Problem Solving Task 2) * There will be multiple 2-hour time slots available for completing the assessment task to cater for students in different time zones.

Summative assessment task 3

Title	**Practical Project Validation Presentation**
Brief description of assessment task	For this assessment, students will be provided with studio activities to have a clear understanding of individual components used in power systems. At the end, the students need to individually demonstrate evidence that they can design a power system. Each student will be required to simulate the build and validate the designed power system based on the learning experience from the design activities during the seminar and understanding of individual components during the studio activities. Students will need to demonstrate the application of OHS understanding, orally explain their project along with an explanation for their design to achieve the desired design objectives. Students will document their results for further consideration in their final report.
Detail of student output	This is an individual task. Each student is required to demonstrate and verbally defend the results of the simulation validation. Students will be assessed on: 1. The construction of their power system designed in assessment 1 2. Their understanding of OHS considerations through verbal communication and the correct use of equipment during the laboratory session (Oral defence) 3. Orally discuss the reason for the demonstrated outcomes
Grading and weighting (% total mark for unit)	30% - marked and graded
This task assesses your achievement of these Unit Learning Outcome(s)	ULO1 - Through student ability to develop solutions for a power engineering design project in a laboratory setting. ULO3 - Through student ability to communicate the working of their project design in a professional manner. ULO4 - Through student ability to demonstrate the OHS standards while validating their project design in a laboratory setting.

This task assesses your achievement of these Graduate Learning Outcome(s)	GLO1 - Through student ability to demonstrate an application of practical skills in building a power system. GLO2 - Through student ability to orally communicate the designed project outcomes to a professional audience. GLO4 - Through student ability to critically analyse power system components and explain the reasons for the demonstrated outcomes. GLO5 - Through student ability to create solutions to the unexpected problems that occur during the practical sessions. GLO6 - Through student ability to work within OHS requirements of the electrical engineering profession.
How and when you will receive feedback on your work	Students will receive ongoing feedback from demonstrators during studio sessions. Students will receive oral feedback after completing the final project validation.
When and how to submit your work	During the last studio session for campus students (Week 10). At the end of the practical session during the intensive week and during online classes in Week 10 for online students.

Summative assessment task 4

Title	**Final Report (group)**
Brief description of assessment task	This assessment task requires students to clearly and professionally communicate the outcomes of their practical experience to design and validate the project. Students are required to provide a written report that evaluates their design and identify potential ethical implications of their design. Students will demonstrate their critical thinking skills to validate the project design based on studio activities and theoretical concepts developed throughout the unit.
Detail of student output	This is a group assessment task. Students will submit a short report of no more than 2500 words. The report will include: 1. Summary of the design 2. The results of the individual components validation 3. The results of the Practical Project Validation 4. How the results compared to the calculations determined in the design 5. Analytical calculations to evaluate the operational characteristics of the designed power system. 6. Suggested recommendations around potential ethical implications including indigenous perspectives.
Grading and weighting (% total mark for unit)	25% - marked and graded *There will be an individualised scaling factor applied based on peer and self-review to reflect your contribution and ability to work well within a team, moderated by the Unit Chair. If this self and peer review is not completed, a penalty of up 25% of the available marks is deducted.*
This task assesses your achievement of these Unit Learning Outcome(s)	ULO1 - Through student ability to apply theoretical understanding of the power engineering concepts to validate the outcomes of the project design. ULO3 - Through student ability to communicate the findings from the project validation in a professional manner.
This task assesses your achievement of these Graduate Learning Outcome(s)	GLO1 - Through student ability to demonstrate their understanding of the key concepts of power engineering design. GLO2 - Through student ability to communicate recommendations to an engineering problem to a professional audience. GLO4 - Through student ability to critically analyse power systems and evaluate the results of practical tests, simulations and calculations.
How and when you will receive feedback on your work	Students are encouraged to seek feedback from their unit chair if required. Students will receive oral feedback during on-campus and online seminars and after completing the final project validation.

When and how to submit your work	Sunday 1 June 2025, end of Week 12 (Study period), 8pm (AEST), via the unit site.

Your learning experiences in this unit

Educator-facilitated (scheduled) learning activities - on-campus unit enrolment

2 x 2 hour seminars per week, 1 x 2 hour practical experience (studio) per week
Educator-facilitated (scheduled) learning activities - online unit enrolment

Online independent and collaborative learning including 1 x 2 hour seminar per week. Students must attend and participate in project-based activities at the Waurn Ponds (Geelong) Campus for the scheduled sessions during the trimester intensive activities as detailed in the unit site.
Typical study commitment

Students will on average spend 300 hours over the teaching period undertaking the teaching, learning and assessment activities for this unit.
This will include educator guided online learning activities within the unit site.
For campus students this time includes weekly online content, designated activities in the seminar and studio sessions, assessment tasks, reading, study time, problem-based exercises and project team meetings. For online students this time should be divided between online learning activities, discussion boards, assessments tasks, readings, study time, project team meetings, and studio sessions during the mandatory 'intensive' week.

There are mandatory on-campus seminar and studio sessions for campus students throughout the trimester. The on-campus attendance requirement for online students during the intensive week will also be announced on the unit site.

Students will engage in learning experiences that include online content, seminars, studio activities and online concepts in the unit site. This unit is a project oriented design based learning (PODBL) experience and therefore students are required to work as a member of a team to learn and complete a power engineering project. The unit site will host unit materials, including recorded lectures and notes, reading materials and assessment requirements. Please note that there is a compulsory campus attendance requirement for all students either at scheduled sessions every week during the trimester (for on campus students) or over two scheduled days during the 'intensive week' program for online students.
Note

At Deakin, courses are delivered within a learning environment that provides all students with equitable and consistent access to facilities, infrastructure, resources, and support to assist student progress and achievement of learning outcomes.

We have introduced new terms to reflect learning activities to enhance your learning experience, aligning with our innovative **DeakinDesign learning principles and practices**. The new terms better reflect how teaching teams will guide you through your learning journey and the types of learning experiences you will have.

'Lectures' are the activities where teaching staff engage you through presentations with student participation.

In **'seminars'**, an educator will guide you in a smaller group of students through highly interactive discussions and activities.

Your units may also include '**practical experiences**' such as '**laboratory**', '**workshops**', '**clinical skills**' and more. These hands-on activities typically take place in specialised facilities with industry tools, equipment or technology to allow you to apply your knowledge practically.

Some other terms

If you see a '**meeting**' in your timetable, this is an optional drop-in session.

'Assessments' or **'team-based learning**' indicate an in-class evaluation of your skills or knowledge. A **'pre-assessment practice'** could be scheduled to prepare you for these assessments.

Find out more

Take a look at the **Learning activities webpage** for a full list of the terminology changes and reasons they were changed.

Unit learning resources

Your unit learning resources can be accessed from your unit site.
The texts and reading list for SEE210 can be found via the University Library.

Note: Select the relevant trimester reading list. Please note that a future teaching period's reading list may not be available until a month prior to the start of that teaching period so you may wish to use the relevant trimester's prior year reading list as a guide only.

Essential learning resources

The unit site, technical/academic/industry facilitator, laboratory facilities, library.

Recommended learning resources

- J. D. Glover, M. S. Sarma, T. J. Overbye, Power System Analysis and Design, 6th Edition (SI), Cengage Learning, 2017.
- R. L. Boylestand, Introductory Circuit Analysis, Any Edition, Merrill Publishing Company/A Bell & Howell Company.
- J. Rosenblatt and M. H. Friedman, Direct and Alternating Current Machinery, Any Edition, McGraw-Hill.
- Fellow students, technical and academic staff, online databases.

The Deakin Software Library provides students with access to software that you may need or find useful for your study at Deakin.

Where to access unit resources

Textbooks can be sourced from various outlets including direct from the publisher, online bookshops, or retailers. Limited copies of textbooks are also available on loan from the University Library.

Key dates for this study period

Trimester 1 teaching period begins	Monday 3 March 2025
Census date	Monday 31 March 2025
Easter/intra-trimester break	Friday 18 April - Sunday 27 April 2025
Trimester 1 teaching period ends	Friday 23 May 2025
Study period (end-of-unit assessment /examination preparation period)	Monday 26 May - Friday 30 May 2025
End-of-unit assessment and examination period	Monday 2 June - Friday 13 June 2025
Inter-trimester break (the period between trimesters)	Monday 16 June - Friday 4 July 2025
Unit results released	Thursday 3 July 2025 (10.30 am)

Unit weekly activities

Week	Commencing	Topic	Special learning activities	Assessment due date
1	3 March 2025	Overview of DC Circuits	Lab Induction	
2#	10 March 2025	Single-Phase AC Circuits	Project Management and Design	
3	17 March 2025	Three-Phase AC Circuits	Project Design and Practical Validation, Ethics	

4	24 March 2025	Transformers	Project Design and Practical Validation	
5	31 March 2025	Transformers	Project Design and Practical Validation	**Assessment 1: Project Design Plan (group)** due Thursday 3 April
6	7 April 2025	Load Characteristics	Project Design and Practical Validation	**Assessment 2: Design Problem Solving Task 1** due Wednesday 9 April
7*~	14 April 2025	Load Characteristics	Project Design and Practical Validation	
8	28 April 2025	Elements of Transmission and Distribution Lines	Project Design and Practical Validation	
9	5 May 2025	Per Unit System	Project Design and Practical Validation	
10	12 May 2025	Basic Power Flow Study	Project Design and Practical Validation	**Assessment 3: Practical Project Validation presentation** due in the last studio session (Campus students), end of practical session during intensive week and during online classes (Online students)
11	19 May 2025	Overall Project Summary	Project Finalisation	
12	27 May 2025		Finalising Project Report	Study period **Assessment 2: Design Problem Solving Task 2** due Wednesday 28 May **Assessment 4: Final Report (group)** due 1 June
13-14^	2 June 2025			End-of-unit assessment and exam period

Labour Day public holiday: **Monday 10 March 2025** - University closed
* Easter/Intra-trimester break: **Friday 18 April - Sunday 27 April 2025** (between weeks 7 and 8)
~ ANZAC Day public holiday: **Friday 25 April 2025** - University closed
^ King's Birthday public holiday: **Monday 9 June 2025** - University closed